VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	23BCE7360

Continuous Assessment Test (CAT) – II - October 2025

Programme	:	B.TECH(CSE)	Semester	:	FALL 2025-26
Course Code & Course Title	:	BCSE355L & Cloud Architecture Design	Class Number	:	CH2025260101063 CH2025260101064 CH2025260101066 CH2025260101067 CH2025260101068 CH2025260101069 CH2025260101071 CH2025260101072
Faculty	:	Dr. Priyaadharshini M Dr. Sandosh Dr. Priyadarshini R Dr. Om Kumar C U Dr. Palani Thanaraj K Dr. Selvam D Dr. Karthika S K Dr. Prabhakaran R	Slot	:	F2+TF2
Duration	:	1 Hour 30 Minutes	Max. Mark		50 Marks

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		A social networking company is scaling to 50 million daily active users with the requirements, • Transactional data: Likes, comments, messages (must be ACID-compliant). • Catalog: Posts, images, and hashtags need low-latency lookups. • Analytics: They want to run real-time dashboards on engagement metrics. • Challenges: Sudden traffic spikes, budget limits for storage, global user base. Propose and illustrate a database solution combining multiple AWS database services. How scalability and cost-effective strategies solves the challenges?	10	CO2	K4
2		Your company runs a popular e-commerce application on AWS using Amazon RDS for its backend database. During festive sales the number of users has doubled and the database is struggling to handle a large number of queries leading to slow response times. At the same time, management is concerned about ensuring that the database remains available and durable not only during an entire	10	CO1	K4

Scanned with OKEN Scanner

		Availability Zone but also in the event of a hardware failure with automatic failover to minimize downtime. As a cloud architect, recommend and describe the configurations suitable to, a. Improve scalability for handling high read traffic with justification. (5 marks) b. Ensure high availability, durability and automatic failover in case of any failure with justification. (5 marks)			
3		You are a cloud administrator for a SaaS platform hosting multiple customer-facing microservices. During a major product launch, the platform experiences sudden spikes in CPU usage, memory consumption, and API errors across compute instances and serverless functions. Your manager asks you to ensure real-time monitoring, automated scaling, and event-driven remediation to maintain platform availability and SLA compliance. a. Design an integrated solution to detect performance anomalies in real time, notify relevant teams, and automatically remediate issues. Illustrate how dashboards, alarms, and automated functions can be combined to respond to spikes efficiently. Include strategies for event filtering and sequential processing of alerts. (5 marks) b. Analyze the challenges of monitoring high-volume, multi-region services and evaluate strategies to ensure efficient metric aggregation, reliable alerting, auto-scaling responsiveness, and cost optimization, while maintaining audit compliance. Include how scaling decisions can be triggered automatically based on monitored metrics. (5 marks)	10	C04	K4
4		A large, global enterprise manages thousands of EC2 instances across multiple AWS accounts, with a mix of production, development, and testing environments. The operations team is struggling with several critical, recurring challenges: • The team needs to ensure all instances are regularly patched with the latest security updates, but the process is manual, error-prone, and difficult to track across a large fleet. • They must enforce strict security configurations. For example, all production EC2 instances must have a specific IAM role attached, and a defined set of ports must be blocked by the security group. • A routine operational task, such as restarting a specific application service on all instances in the us-east-1 region, needs to be automated and executed from a central management account. Propose a unified, automated, and compliant framework to address	10	CO2	K6

		these challenges. Justify how this solution addresses the complexities of a large, multi-account environment.			
5		A fintech startup is migrating its customer-facing web application to AWS. The application handles sensitive financial data and must comply with strict data protection regulations. The team decides to use Amazon EC2 for compute, Amazon S3 for storage, and AWS Identity and Access Management (IAM) for access control. During a security audit, it is discovered that the S3 buckets are publicly accessible and some EC2 instances are missing critical OS patches. Based on the AWS Shared Responsibility Model, assess how security responsibilities are divided and managed between AWS and the startup in this scenario. Identify which security lapses fall under the startup's domain and explain how they should address them. Support your answer with specific responsibilities from the model.	10	CO3	K4

**********All the best **********

Faculty Signature

Scanned with OKEN Scanner